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1. Klausur PraMa 2

Gruppe B

Aufgabe 1:

$$a) \quad \Theta \in [0, \pi], \varphi \in [0, \frac{\pi}{2}]$$

$$x^2 + y^2 + z^2 = 9$$

$$r^2 = 9 \Rightarrow r = 3$$

$$r(\Theta, \varphi) = \begin{pmatrix} 3 \sin \Theta \cos \varphi \\ 3 \sin \Theta \sin \varphi \\ 3 \cos \Theta \end{pmatrix}$$

$$\frac{\partial r}{\partial \Theta} = \begin{pmatrix} 3 \cos \Theta \cos \varphi \\ 3 \cos \Theta \sin \varphi \\ -3 \sin \Theta \end{pmatrix}, \quad \frac{\partial r}{\partial \varphi} = \begin{pmatrix} -3 \sin \Theta \sin \varphi \\ 3 \sin \Theta \cos \varphi \\ 0 \end{pmatrix}$$

$$n(\Theta, \varphi) = \begin{pmatrix} 3 \cos \Theta \cos \varphi \\ 3 \cos \Theta \sin \varphi \\ -3 \sin \Theta \end{pmatrix} \times \begin{pmatrix} -3 \sin \Theta \sin \varphi \\ 3 \sin \Theta \cos \varphi \\ 0 \end{pmatrix}$$

$$= \begin{pmatrix} 0 - (-9 \sin^2 \Theta \cos \varphi) \\ - (0 - (9 \sin^2 \Theta \sin \varphi)) \\ 9 \cos \Theta \sin \Theta \cos^2 \varphi - (-9 \cos \Theta \sin \Theta \sin^2 \varphi) \end{pmatrix}$$

$$n(\Theta, \varphi) = \begin{pmatrix} 9 \sin^2 \Theta \cos \varphi \\ 9 \sin^2 \Theta \sin \varphi \\ 9 \cos \Theta \sin \Theta \end{pmatrix}$$

$$n(\Theta, \varphi) = 9 \cdot \sin \Theta \cdot \begin{pmatrix} \sin \Theta \cos \varphi \\ \sin \Theta \sin \varphi \\ \cos \Theta \end{pmatrix} \quad \checkmark$$

$$b) \quad W(x, y, z) = \begin{pmatrix} \frac{y}{\sqrt{x^2 + y^2}} \\ \frac{x}{\sqrt{x^2 + y^2}} \\ 0 \end{pmatrix}$$

$$r(\Theta, \varphi) = \begin{pmatrix} 3 \sin \Theta \cos \varphi \\ 3 \sin \Theta \sin \varphi \\ 3 \cos \Theta \end{pmatrix}$$

$$W(r(\Theta, \varphi)) = \begin{pmatrix} \frac{3 \sin \Theta \sin \varphi}{\sqrt{9 \sin^2 \Theta \cos^2 \varphi + 9 \sin^2 \Theta \sin^2 \varphi}} \\ \frac{3 \sin \Theta \cos \varphi}{\sqrt{9 \sin^2 \Theta \cos^2 \varphi + 9 \sin^2 \Theta \sin^2 \varphi}} \\ 0 \end{pmatrix}$$

$$= \begin{pmatrix} \frac{3 \sin \Theta \sin \varphi}{3 \sin \Theta} \\ \frac{3 \sin \Theta \cos \varphi}{3 \sin \Theta} \\ 0 \end{pmatrix}$$

$$W(r(\theta, \varphi)) = \begin{pmatrix} \sin \varphi \\ \cos \varphi \\ 0 \end{pmatrix}$$

$$\int_0^{\frac{\pi}{2}} \int_0^{\pi} \begin{pmatrix} \sin \varphi \\ \cos \varphi \\ 0 \end{pmatrix} \cdot \begin{pmatrix} 9 \sin^2 \theta \cos \varphi \\ 9 \sin^2 \theta \sin \varphi \\ 9 \cos \theta \sin \theta \end{pmatrix} d\theta d\varphi =$$

$$\int_0^{\frac{\pi}{2}} \int_0^{\pi} 9 \sin^4 \theta \sin \varphi \cos \varphi + 9 \sin^4 \theta \sin \varphi \cos \varphi d\theta d\varphi =$$

$$18 \cdot \int_0^{\frac{\pi}{2}} \int_0^{\pi} \sin^4 \theta \sin \varphi \cos \varphi d\theta d\varphi$$

$$[\text{Hinweis: } \int_0^{\frac{\pi}{2}} \sin^4 \theta d\theta = \frac{\pi}{2}]$$

$$9\pi \cdot \int_0^{\frac{\pi}{2}} \sin \varphi \cos \varphi d\varphi$$

NR:

$$\int \sin \varphi \cos \varphi d\varphi = \sin^2 \varphi - \int \sin \varphi \cos \varphi d\varphi$$

$$\sin \varphi' = \cos \varphi$$

$$\int \cos \varphi d\varphi = \sin \varphi$$

$$2. \int \sin \varphi \cos \varphi d\varphi = \sin^2 \varphi$$

$$\int \sin \varphi \cos \varphi d\varphi = \frac{1}{2} \sin^2 \varphi$$

$$9\pi \cdot \frac{1}{2} \cdot [\sin^2 \varphi]_0^{\frac{\pi}{2}} =$$

$$= \frac{9}{2} \pi$$

$$c) \text{rot}(U) = \begin{pmatrix} \frac{y}{\sqrt{x^2+y^2}} \\ -\frac{x}{\sqrt{x^2+y^2}} \\ 0 \end{pmatrix}$$

$$x^2 + y^2 \leq 4 \Rightarrow R = 2$$

$$-1 \leq z \leq 1 \quad z \in [-1, 1], \varphi \in [0, 2\pi]$$

$$r(\varphi, z) = \begin{pmatrix} R \cos \varphi \\ R \sin \varphi \\ z \end{pmatrix}$$

$$\frac{\partial r}{\partial \varphi} = \begin{pmatrix} -R \sin \varphi \\ R \cos \varphi \\ 0 \end{pmatrix}, \quad \frac{\partial r}{\partial z} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

$$n(\varphi, z) = \begin{pmatrix} -R \sin \varphi \\ R \cos \varphi \\ 0 \end{pmatrix} \times \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} R \cos \varphi \\ R \sin \varphi \\ 0 \end{pmatrix}$$

$$\text{rot}(U(r)) = \begin{pmatrix} \frac{R \sin \varphi}{\sqrt{R^2 \cos^2 \varphi + R^2 \sin^2 \varphi}} \\ -\frac{R \cos \varphi}{\sqrt{R^2 \cos^2 \varphi + R^2 \sin^2 \varphi}} \\ 0 \end{pmatrix}$$

$$\text{rot}(U(r)) = \begin{pmatrix} \sin \varphi \\ -\cos \varphi \\ 0 \end{pmatrix}$$

$$\int_0^{2\pi} \int_{-1}^1 \begin{pmatrix} \sin \varphi \\ -\cos \varphi \\ 0 \end{pmatrix} \cdot \begin{pmatrix} R \cos \varphi \\ R \sin \varphi \\ 0 \end{pmatrix} dz d\varphi =$$

$$\int_0^{2\pi} \int_{-1}^1 R \sin \varphi \cos \varphi - R \sin \varphi \cos \varphi dz d\varphi$$

$$\int \text{rot}(U) dS = 0$$

Aufgabe 2:

Daniel Köstler

a)

$$\frac{\partial \Phi}{\partial x} = \ln(z) \cos(x+z) - y^2 e^{x+z}$$

$$\Phi = \int \ln(z) \cos(x+z) - y^2 e^{x+z} dx = \left| \begin{array}{l} u = x+z \\ du = dx \end{array} \right|$$

$$= \int \ln(z) \cdot \cos(u) du - \int y^2 e^u du$$

$$= \ln(z) \sin(u) - y^2 e^u + f(y, z)$$

$$\Phi = \ln(z) \sin(x+z) - y^2 e^{x+z} + f(y, z)$$

$$\frac{\partial \Phi}{\partial y} = -2ye^{x+z} + f'(y, z)$$

$$f'(y, z) \stackrel{!}{=} - \frac{z}{1+(yz)^2}$$

NR:

$$\int - \frac{z}{1+(yz)^2} dy = \left| \begin{array}{l} u = yz \\ du = z dy \\ \frac{du}{z} = dy \end{array} \right| = - \int \frac{1}{1+u^2} du$$

$$\begin{aligned} & [\text{Tipp: } \int \frac{1}{1+u^2} du = \arctan(u)] \\ & = -\arctan(yz) + c \end{aligned}$$

$$f(y, z) = -\arctan(yz) + g(z)$$

Daniel Köstler

$$\Phi = \ln(z) \sin(x+z) - y^2 e^{x+z} - \arctan(yz) + g(z)$$

$$\frac{\partial \Phi}{\partial z} = \frac{1}{z} \sin(x+z) + \ln(z) \cos(x+z) - y^2 e^{x+z} - \frac{y}{1+(yz)^2} + g'(z)$$

NR:

$$\frac{\partial}{\partial z} (\ln(z) \sin(x+z)) = \ln(z) \cos(x+z) + \frac{1}{z} \sin(x+z)$$

$$f_1 = \ln(z), f_1' = \frac{1}{z}$$

$$f_2 = \sin(x+z), f_2' = \cos(x+z)$$

$$\frac{\partial}{\partial z} (-y^2 e^{x+z}) = -y^2 e^{x+z}$$

$$\frac{\partial}{\partial z} (-\arctan(yz)) = -\frac{y}{1+(yz)^2}$$

$$g'(z) \stackrel{!}{=} 0$$

$$g(z) = c$$

$$\Phi = \ln(z) \sin(x+z) - y^2 e^{x+z} - \arctan(yz) + c$$

$$\Phi(1, 0, 1) = 0 - 0 - 0 + c = 1$$

$$c \stackrel{!}{=} 1$$

$$\Phi(x, y, z) = \ln(z) \sin(x+z) - y^2 e^{x+z} - \arctan(yz) + 1$$

b) Das Potential ist in dem Bereich definiert in dem gilt:

- $1 + (yz)^2 \neq 0$
- $z \neq 0$

c) Da ein Potential existiert, hängt das Ergebnis des Kurvenintegrals nur von Anfangs- und Endpunkt ab. (wegunabhängig)

Weiters existiert ein Parameterwechsel, somit sind die Kurven äquivalent.

Somit werden sich die Kurvenintegrale nicht unterscheiden!

Aufgabe 3:

Daniel Köstler

$$F(x, y, z) = \begin{pmatrix} x+4y \\ 3y-x \\ 2x+\sqrt{z} \end{pmatrix}$$

$$\sqrt{x^2+y^2} \leq 2z, \quad 0 \leq z \leq 1$$

$$r \leq 2z, \quad \frac{r}{2} \leq z$$

$$r \leq 2, \quad r \in [0, 2]$$

[Durchlaufrichtung so,
dass Normalvektor von
Fläche wegzeigt
(positiv)]

$$a) \quad r(r, \varphi) = \begin{pmatrix} r \cdot \cos \varphi \\ r \cdot \sin \varphi \\ \frac{r}{2} \end{pmatrix}$$

$$\frac{\partial r}{\partial r} = \begin{pmatrix} \cos \varphi \\ \sin \varphi \\ \frac{1}{2} \end{pmatrix}, \quad \frac{\partial r}{\partial \varphi} = \begin{pmatrix} -r \sin \varphi \\ r \cos \varphi \\ 0 \end{pmatrix}$$

$$n(r, \varphi) = \begin{pmatrix} \cos \varphi \\ \sin \varphi \\ \frac{1}{2} \end{pmatrix} \times \begin{pmatrix} -r \sin \varphi \\ r \cos \varphi \\ 0 \end{pmatrix}$$

$$= \begin{pmatrix} 0 - \frac{1}{2} r \cos \varphi \\ -(0 - (-\frac{1}{2} r \sin \varphi)) \\ r \cos^2 \varphi + r \sin^2 \varphi \end{pmatrix}$$

$$n(r, \varphi) = \begin{pmatrix} -\frac{1}{2} r \cos \varphi \\ -\frac{1}{2} r \sin \varphi \\ r \end{pmatrix}$$

setze: $r=1, \varphi=0$

$$n(1,0) = \begin{pmatrix} -\frac{1}{2} \\ 0 \\ 1 \end{pmatrix} \quad \text{⚡ negativ orientiert}$$

$$n(r,\varphi) = \begin{pmatrix} \frac{1}{2}r\cos\varphi \\ \frac{1}{2}r\sin\varphi \\ -r \end{pmatrix} \quad \checkmark \text{ positiv orientiert}$$

b)

$$\begin{pmatrix} \frac{\partial}{\partial x} \\ \frac{\partial}{\partial y} \\ \frac{\partial}{\partial z} \end{pmatrix} \cdot \begin{pmatrix} x+4y \\ 3y-x \\ 2x+\sqrt{z} \end{pmatrix} = 1+3+\frac{1}{2}z^{-\frac{1}{2}}$$

$$\int 4 + \frac{1}{2\sqrt{z}} dV = \int_0^{2\pi} \int_0^2 \int_0^1 \left(4r + \frac{r}{2\sqrt{z}} \right) dz dr d\varphi$$

$$= 8\pi \cdot \int_0^2 r dr + \pi \cdot \int_0^2 \int_0^1 \frac{r}{\sqrt{z}} dz dr$$

NR:

$$\int z^{-\frac{1}{2}} dz = \frac{z^{\frac{1}{2}}}{\frac{1}{2}} = 2z^{\frac{1}{2}} = 2\sqrt{z}$$

$$= 8\pi \cdot \left[\frac{r^2}{2} \right]_0^2 + \pi \cdot \left[2\sqrt{z} \right]_0^1 \cdot \left[\frac{r^2}{2} \right]_0^2$$

$$= 8\pi \cdot 2 + \pi \cdot 2 \cdot 2 = 16\pi + 4\pi = 20\pi$$

$$\int \operatorname{div}(F) dV = 20\pi$$

$$\begin{aligned} \sin(x)' &= \cos(x) & \int \sin(x) dx &= -\cos(x) \\ \cos(x)' &= -\sin(x) & \int \cos(x) dx &= \sin(x) \end{aligned}$$

$$\begin{aligned} \int \sqrt{1+x^2} &\Rightarrow x = \sinh(t) \\ \int \sqrt{x^2-1} &\Rightarrow x = \cosh(t) \\ \int \sqrt{1-x^2} &\Rightarrow x = \cos(t)/\sin(t) \\ \cosh^2(t) - \sinh^2(t) &= 1 \end{aligned}$$

$$\begin{aligned} \int_0^{2\pi} \sin^2(x) dx &= \pi \\ \int_0^{2\pi} \cos^2(x) dx &= \pi \end{aligned}$$

$$\begin{aligned} \sin^2(x) &= \frac{1}{2} \cdot (1 - \cos(2x)) \\ \sin(x)\sin(y) &= \frac{1}{2} \cdot (\cos(y-x) - \cos(y+x)) \\ \cos(x)\cos(y) &= \frac{1}{2} \cdot (\cos(y+x) + \cos(y-x)) \\ \sin(x)\cos(y) &= \frac{1}{2} \cdot (\sin(y+x) - \sin(y-x)) \end{aligned}$$

$$1 - 2\sin^2(x) = \cos(2x)$$

Satz von Green:

z.B.: $\int_C x dx + y dy \Rightarrow a = \begin{pmatrix} y \\ x \end{pmatrix}$

Normalvektor in $\mathbb{R}^2$: $r'(t) = \begin{pmatrix} x'(t) \\ y'(t) \end{pmatrix} \Rightarrow n = \begin{pmatrix} -y'(t) \\ x'(t) \end{pmatrix}$

Radius bei Ellipsoid:

$$\left(\frac{x}{a}\right)^2 + \left(\frac{y}{b}\right)^2 + \left(\frac{z}{c}\right)^2 = 1 \Rightarrow \left(\frac{x}{a}\right)^2 + \left(\frac{y}{b}\right)^2 = r^2$$

φ von $x \rightarrow y$
 Θ von $z \rightarrow y$

Zylinderkoordinaten: $x = r \cdot \cos \varphi$, $y = r \cdot \sin \varphi$, $z = z$

Kugelkoordinaten: $x = R \cdot \sin \Theta \cdot \cos \varphi$, $y = R \cdot \sin \Theta \cdot \sin \varphi$, $z = R \cdot \cos \Theta$

Volumen:	Oberfläche:	Volumen Ellipsoid:
Zylinder: $r^2 \pi \cdot h$	$2\pi r h$	$\frac{4}{3} \pi a b c$
Kugel: $\frac{4\pi r^3}{3}$	$4\pi r^2$	

$$\begin{aligned} \int f(x)g(x) dx &= f(x) \cdot d(x) - \int c(x) d(x) dx \\ f'(x) &= c(x) \\ \int g(x) dx &= d(x) \end{aligned}$$

Fläche Ellipse:
 $\pi \cdot a \cdot b$

$$\begin{aligned} (f(x)g(x))' &= f'(x)g(x) + f(x)g'(x) \\ \left(\frac{f(x)}{g(x)}\right)' &= \frac{f'(x)g(x) - f(x)g'(x)}{g(x)^2} \end{aligned}$$

Wenn bei Satz von Stokes falscher $n \rightarrow [2\pi, 0]$ bei Rand